

Cs301 MidTerm Object File:01

(Lectures from 1 to 23)

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Question No: 1 (Marks: 1) - Please choose one

A queue where the de-queue operation depends not on FIFO, is called a priority queue

- ▶ False
- ▶ **True (Page 101)**

Question No: 2 (Marks: 1) - Please choose one

The data of the problem is of 2GB and the hard disk is of 1GB capacity, to solve this problem we should

- ▶ Use better data structures
- ▶ **Increase the hard disk space (Page 5)**
- ▶ Use the better algorithm
- ▶ Use as much data as we can store on the hard disk

Question No: 3 (Marks: 1) - Please choose one

Consider the function X as under

```
int X (int& Value)
{
    return Value;
}
```

Now a and b are integers in a calling function. Which one of the following is a valid call to the above function X.

- ▶ a = X (b) ;
- ▶ **a = X (&b) ;**
- ▶ a = X (*b) ;
- ▶ None of the given options

Here function argument passing by reference method is used, so when we call a function we will give the variable reference as parameter.

Question No: 4 (Marks: 1) - Please choose one

In the call by value methodology, a copy of the object is passed to the called function.

- ▶ False
- ▶ **True (Page 202)**

Question No: 5 (Marks: 1) - Please choose one

The tree data structure is a

- ▶ Linear data structure
- ▶ **Non-linear data structure (Page 112)**
- ▶ Graphical data structure
- ▶ Data structure like queue

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Question No: 6 (Marks: 1) - Please choose one

When should you use a **const reference** parameter?

- ▶ Whenever the parameter has huge size.
- ▶ **Whenever the parameter has huge size, the function changes the parameter within its body, and you do NOT want these changes to alter the actual argument.**
- ▶ Whenever the parameter has huge size, the function changes the parameter within its body, and you DO want these changes to alter the actual argument.
- ▶ Whenever the parameter has huge size, and the function does not change the parameter within its body.

Declaring a parameter as a const simply means that the function can't change the value of its parameters.

Question No: 7 (Marks: 1) - Please choose one

Here is the start of a C++ class declaration:

```
class foo
{
public:
    void x(foo f);
    void y(const foo f);
    void z(foo f) const;
    ...
}
```

Which of the three member functions can alter the PRIVATE member variables of the foo object that activates the function?

- ▶ Only x can alter the private member variables of the object that activates the function.
- ▶ Only y can alter the private member variables of the object that activates the function.
- ▶ Only z can alter the private member variables of the object that activates the function.
- ▶ **Two of the functions can alter the private member variables of the object that activates the function.**

Only the x and y can alter the private member variable of the foo class object. Last Option is more correct but not exact. In the last option the two function name are not mentioned

Question No: 8 (Marks: 1) - Please choose one

What is the maximum depth of recursive calls a function may make?

- ▶ 1
- ▶ 2
- ▶ n (where n is the argument)
- ▶ **There is no fixed maximum**

Question No: 9 (Marks: 1) - Please choose one

Suppose n is the number of nodes in a complete Binary Tree then maximum steps required for a search operation are,

- ▶ **$\log_2(n+1) - 1$ (Page 139)**
- ▶ $\log_2(n+1)$
- ▶ $\log_2(n) - 1$
- ▶ $\log_2(n)$

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Question No: 10 (Marks: 1) - Please choose one

In the linked list implementation of the stack class, where does the push member function places the new entry on the linked list?

- ▶ **At the head** (Page 53)
- ▶ At the tail
- ▶ After all other entries that are greater than the new entry.
- ▶ After all other entries that are smaller than the new entry.

Question No: 11 (Marks: 1) - Please choose one

Suppose we have a *circular* array implementation of the queue class, with ten items in the queue stored at data[2] through data[11]. The CAPACITY is 42, i.e., the array has been declared to be of size 42. Where does the push member function place the new entry in the array?

- ▶ data[1]
- ▶ data[2]
- ▶ data[11]
- ▶ **data[12]**

Question No: 12 (Marks: 1) - Please choose one

The expression $AB+C^*$ is called?

- ▶ Prefix expression
- ▶ **Postfix expression** (Page 70)
- ▶ Infix expression
- ▶ None of these

Question No: 13 (Marks: 1) - Please choose one

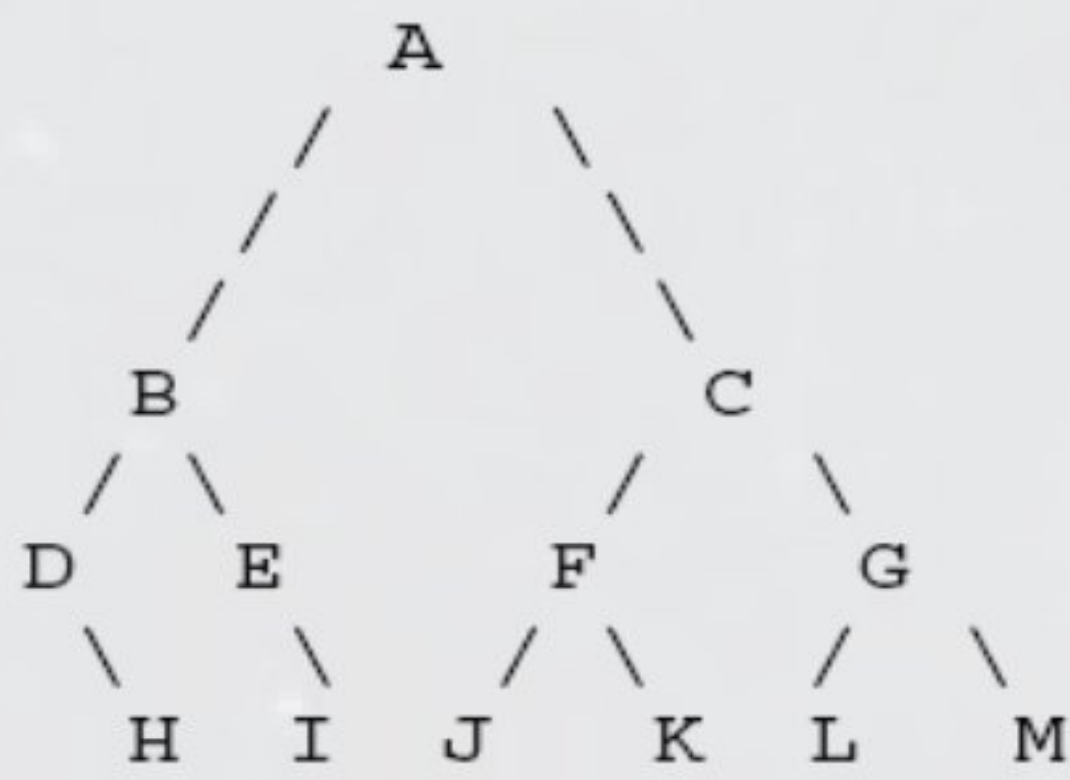
_____ is a binary tree where every node has a value, every node's left subtree contains only values less than or equal to the node's value, and every node's right subtree contains only values that are greater than or equal?

- ▶ Strictly Binary Tree
- ▶ **Binary Search tree** [Click here for detail](#)
- ▶ AVL tree
- ▶ All of these

Question No: 14 (Marks: 1) - Please choose one

Consider the following binary search tree (BST):

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If node A in the BST is deleted, which two nodes are the candidates to take its place?

- ▶ J and I
- ▶ H and E
- ▶ D and E
- ▶ **L and M**

Question No: 15 (Marks: 1) - Please choose one

Let's call the node as that requires re-balancing. Consider the two cases given below:

- 1) An insertion into left sub tree of the left child of a
- 2) An insertion into right sub tree of the right child of a.

Which of the following statement is correct about these two cases?

- 1) The insertion occurs outside (i.e., left-left or right-right) in cases 1 and 2 single rotation can fix the balance in these two cases.
- 2) The insertion occurs inside ((i.e., left-left or right-right) in cases 1 and 2. single rotation cannot fix the balance in these two cases

Question No: 16 (Marks: 1) - Please choose one

We access elements in AVL Tree in,

- ▶ Linear way only
- ▶ **Non Linear way only**
- ▶ Both linear and non linear ways
- ▶ None of the given options.

Question No: 17 (Marks: 2)

AVL Tree is,

- ▶ **Non Linear data structure** [Click here for detail](#)
- ▶ Linear data structure
- ▶ Hybrid data structure (Mixture of Linear and Non Linear)
- ▶ None of the given options.